21EIE506T - MODEL BASED DEVELOPMENT OF CYBER-PHYSICAL SYSTEMS

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Course Learning Rationale (CLR):	*The purpose of learning this course is to:*
CLR-1 :	*Introduce the basic concepts of cyber-physical system and modeling of a continuous system*
CLR-2 :	*Know and understand the discrete modeling of a system*
CLR-3 :	*Impart the adequate information about hybrid system and state machines*
CLR-4 :	*Understand the embedded systems design*
CLR-5 :	*Impart the knowledge about verification and validation of embedded system design*
CLR-6 :	*Understand the characteristics of cyber-physical systems*

Course Learning Outcomes (CLO):	*At the end of this course, learners will be able to:*
CLO-1 :	*Infer the basic concepts of cyber physical systems and modeling in continuous domain*
CLO-2 :	*Develop the discrete model for continuous system*
CLO-3 :	*Formulate the hybrid system and its interactions*
CLO-4 :	*Design and develop the embedded system*
CLO-5 :	*Validate the embedded system design for specific applications*
CLO-6 :	*Assess the characteristics of cyber-physical systems*

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Introduction

- Over the past decades, science and technology have gone through tremendous changes in terms of computing, communications, and control to provide a wide range of applications in all domains.

- This advancement provides opportunities to bridge physical components and cyber space, leading to cyber-physical systems (CPS).

- The notion of CPS is to use computing (sensing, analyzing, predicting, understanding), communication (interaction, intervene, interface management), and control (interoperate, evolve, evidence-based certification) to make intelligent and autonomous systems.

- The complexity in CPS has increased due to the integration of cyber components with physical systems.

- A comprehensive knowledge base of the CPS domain is required not only for researchers and practitioners but also for policy makers and system managers.

Cyber-Physical System

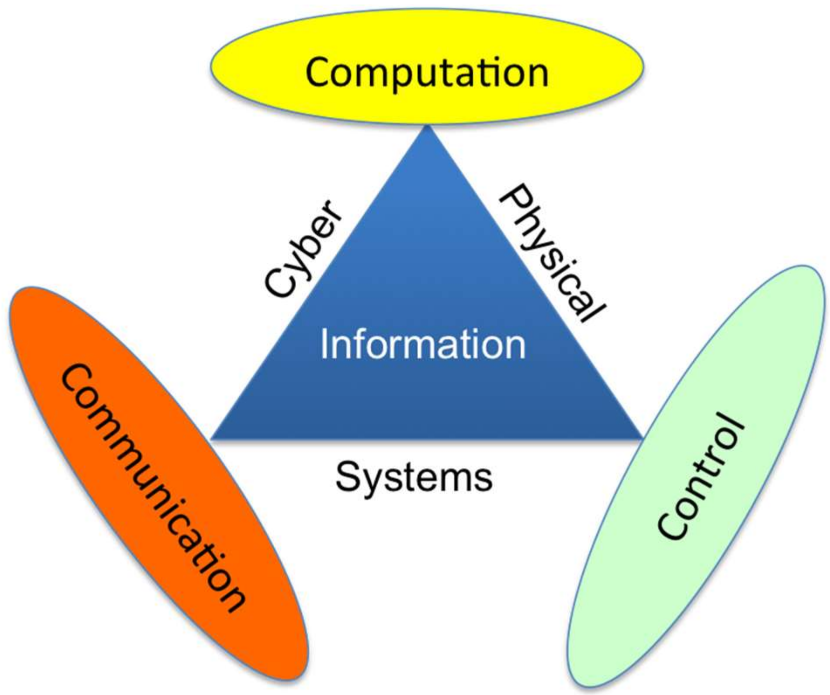


Figure: Building Blocks

- A cyber-physical system is a system that combines physical and computer or cyber components.
- The physical component consists of systems that exist in nature, such as biological entities as well as those developed by humans, such as transportation and energy-producing systems.
- This component exists, operates, and interacts with its environment in continuous or ordinary time.
- The cyber component consists of systems and entities involved in processing, communicating, and controlling information via computational means.

- The term cyber-physical systems was coined in 2006 by the United States National Science Foundation (NSF) program manager Dr.Helen Gill .

- However, these systems have a much longer history that dates back to the beginning of cybernetics, which was defined by mathematician Norbert Wiener as the science of control and communication in machines and humans.

- Cyber-physical systems (CPS) combine, and build on, elements from different scientific theories and engineering disciplines, including cybernetics, embedded systems, distributed control, sensor networks, control theory and systems engineering.

- These theories and disciplines support the optimal integration of digital and physical components, which maximizes the synergies between the two and allows achieving significant benefits both in terms of cost, performance and overall life- cycle sustainability.

Definition

- A cyber-physical system is a collection of interconnected computing devices interacting with the physical world. The computing devices together constitute a cyber system that regulates the behavior of the physical world.

- The cyber system closely monitors the physical world through sensors, computes required control laws based on the current state of the physical world, and applies the computed control law to the physical world through actuators. The sensors, the controllers, and the actuators are developed on top of an embedded platform.

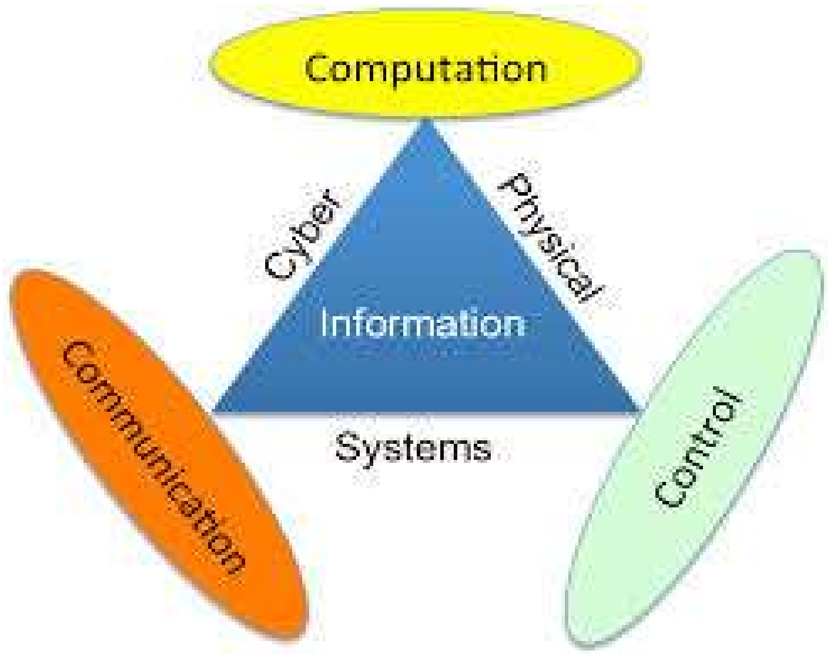
- Thus, the cyber component of a cyber-physical system is often termed as an embedded control system.

- Developing an embedded control system requires the understanding of the physical world with which the system has to interact. The understanding of the physical world is captured in a faithful model that is used for synthesizing feedback control laws using control theoretic methods.

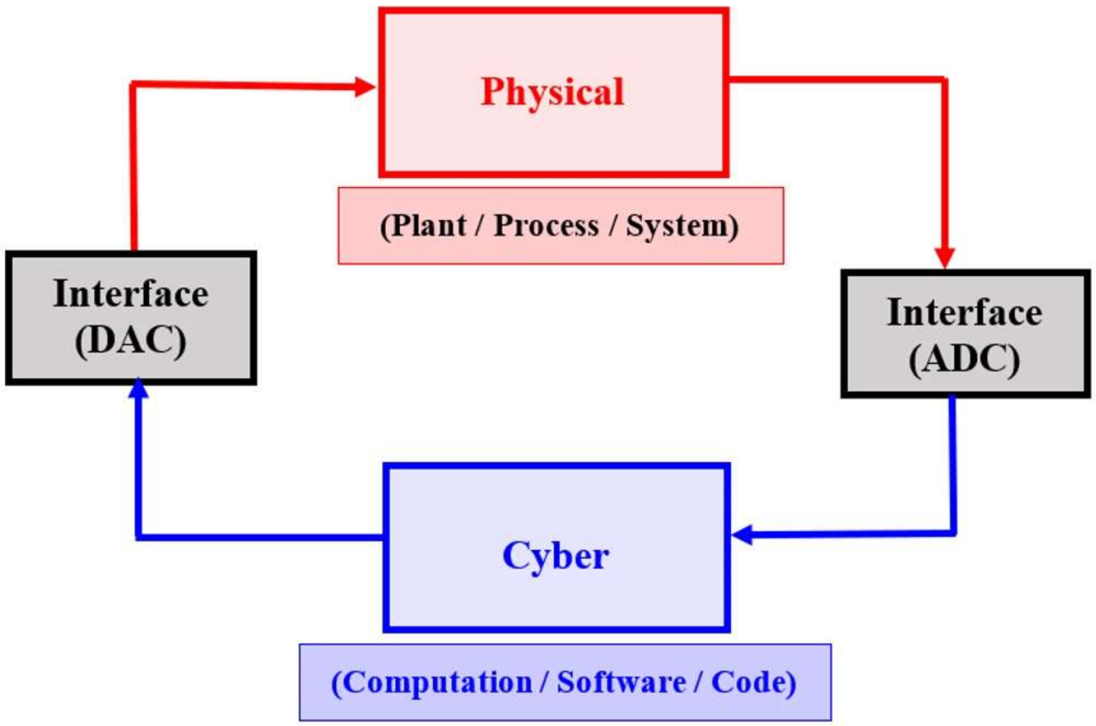
- Implementing the feedback control law on the embedded computing platform requires addressing the challenges of embedded computing, for example, the availability of limited resources in terms of computing power and memory, stringent timing requirements, and so on.

- Moreover, most cyber-physical systems are safety-critical. Thus, it is essential that the correctness of such systems is established through the use of formal verification techniques.

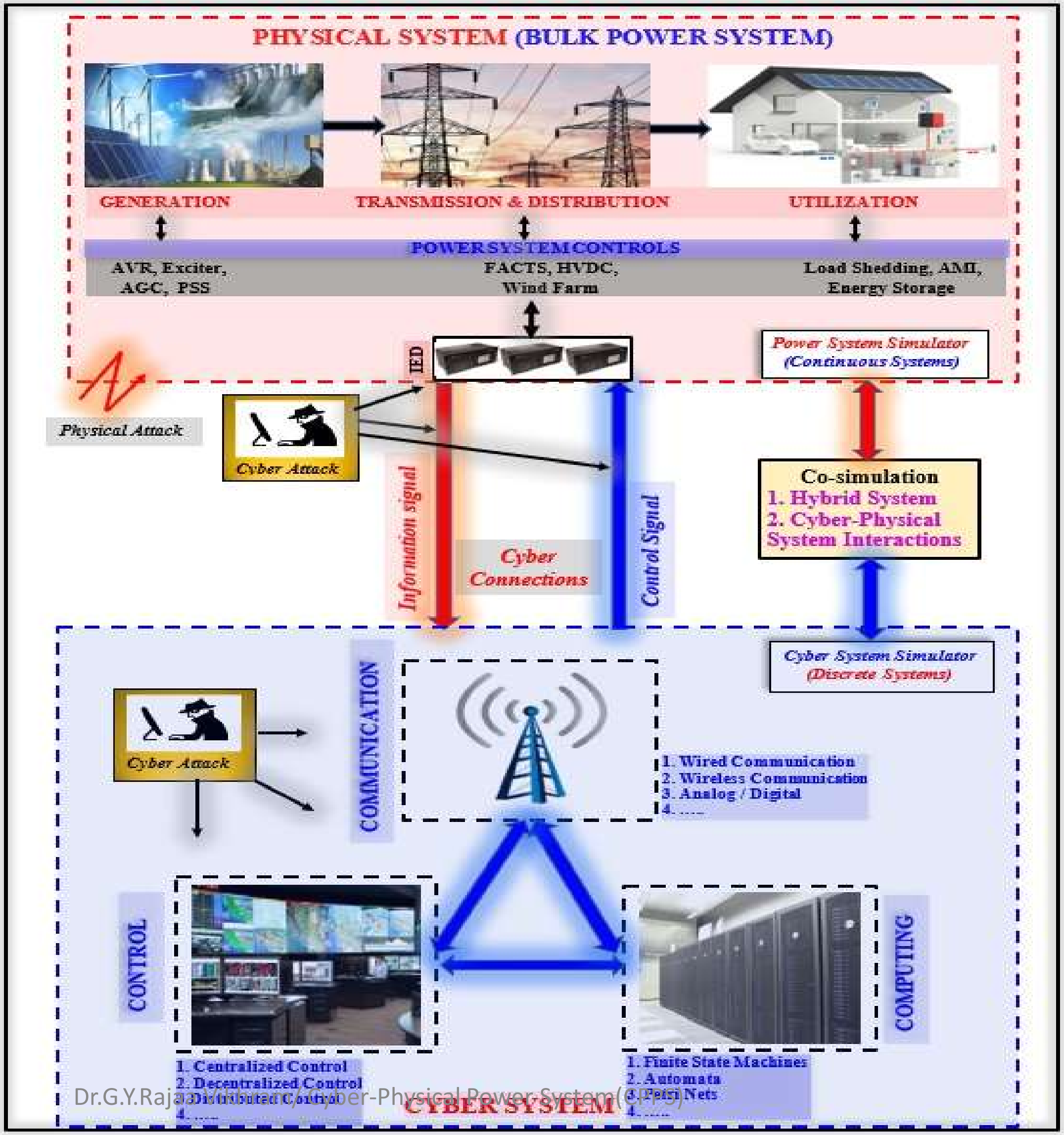
- CPS requires three fundamental attributes to be present, also known as the Three Cs – Communication, Control and Computing. Unless these three elements are present, you will not have a system where physical processes can affect computations and vice versa.



Structure of CPS



Structure of CPPS



Defining CPS and CPPS

- **A Cyber-Physical System (CPS)** is a smart networked system that integrates computing, control, sensing and networking with physical processes.

- **Cyber-Physical Power System (CPPS)** combines computation, communication and control technologies with a physical power system.

- Interactions and interdependencies between physical components (generators, lines, transformers, etc.) and cyber components (control and metering devices, intelligent automation, communication links etc.) are becoming more complex.

The events in CPPS can be grouped into:

1) events in the physical grid (components, grid topology)

2) events in the cyber infrastructure (software and data communication)

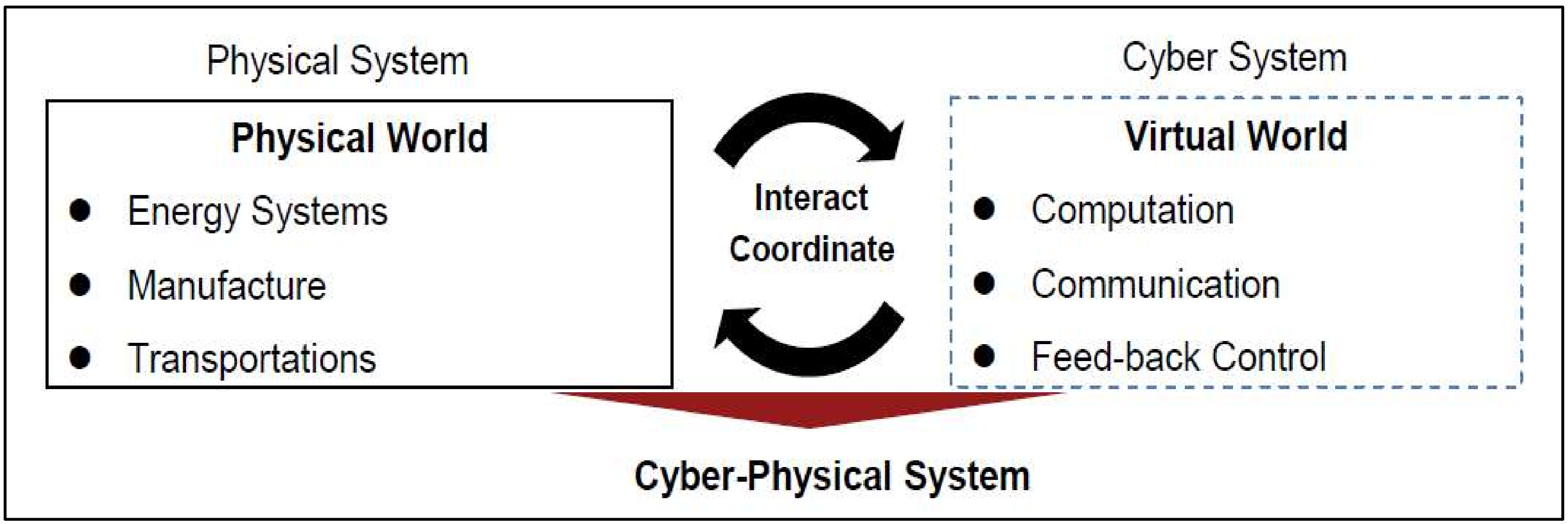
3) dependent events in cyber and physical power system components (control systems, state estimation)

Characteristics of Physical and Cyber System in CPPS

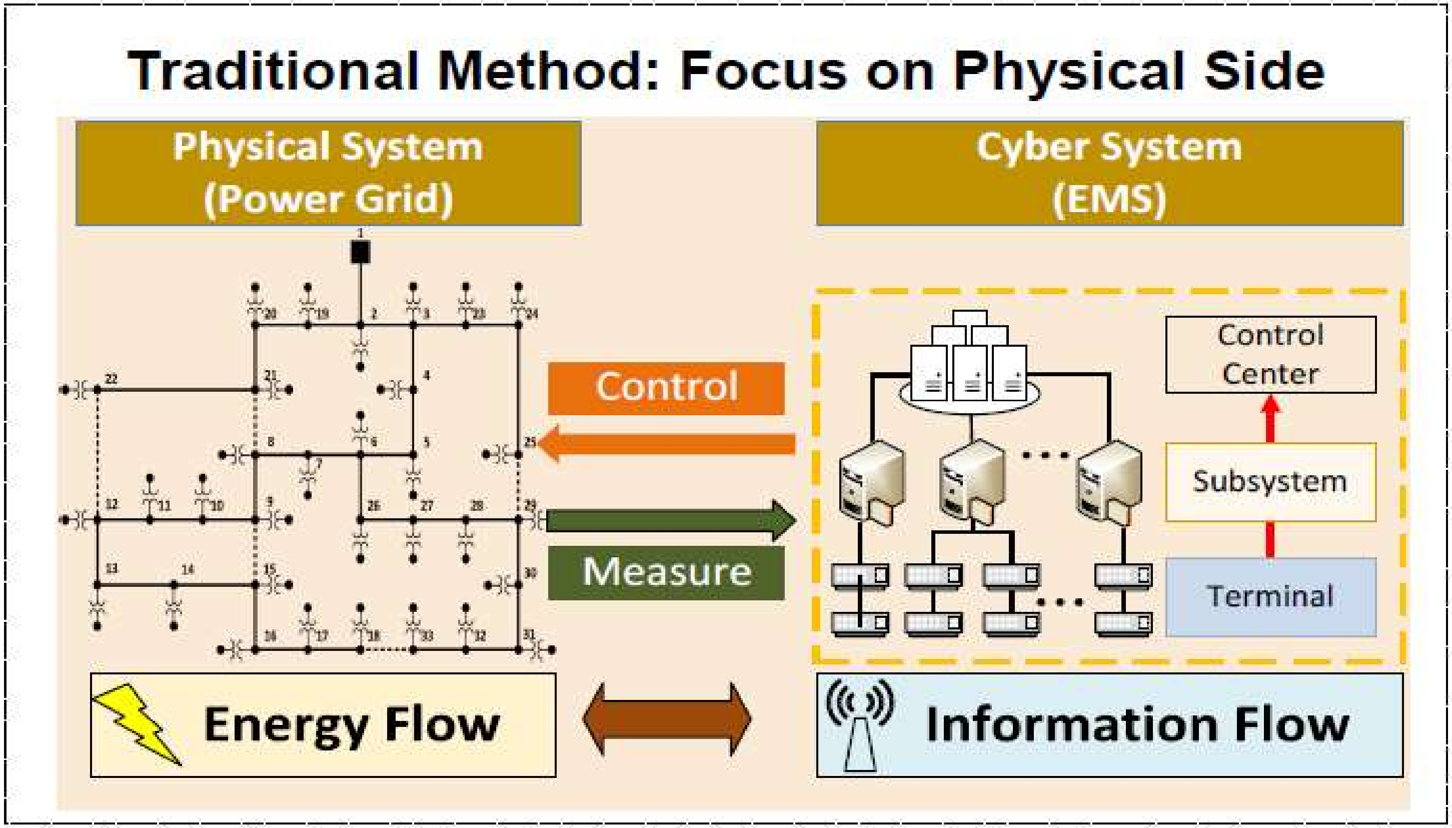
S. No	Characteristics	Physical System	Cyber System
1.	Components	Generator, Transformer, Transmission Line, Circuit Breaker, Protective Relay, Load, etc.	Control Systems, Computing Devices, Communication Networks
2.	Nature of System	Continuous, Dynamic behaviour	Discrete, Static behaviour
3.	Modelling	Differential-Algebraic Equations	Difference Equations
4.	System State	Energy Flow	Information Flow
5.	Branch Model	Power Grid Model – Energy Generation, Energy Transmission and Energy Distribution	Information Flow Oriented Model – Data Transmission, Data Processing, and Data Pool
6.	Condition	Generation and Load Balance, Power Transmission Limits	Interdependent operation balance among Control, Computing, and Communication functions
7.	Contingency	Physical Contingency.	Cyber Contingency
8.	Types of Contingency	Line Fault, Generator Outage, Load Outage, Environmental effects, etc.	Cyber Attacks, Communication Latency, Malicious Control effects, etc.
9.	Stability & Security	Power System Stability & Power System Security	Networked Control System Stability & Cyber Security
10.	Event Synchronization	Asynchronous	Synchronous

Core Idea

- By intertwining physical and software components, CPS presents a higher combination and coordination between physical and computational elements.



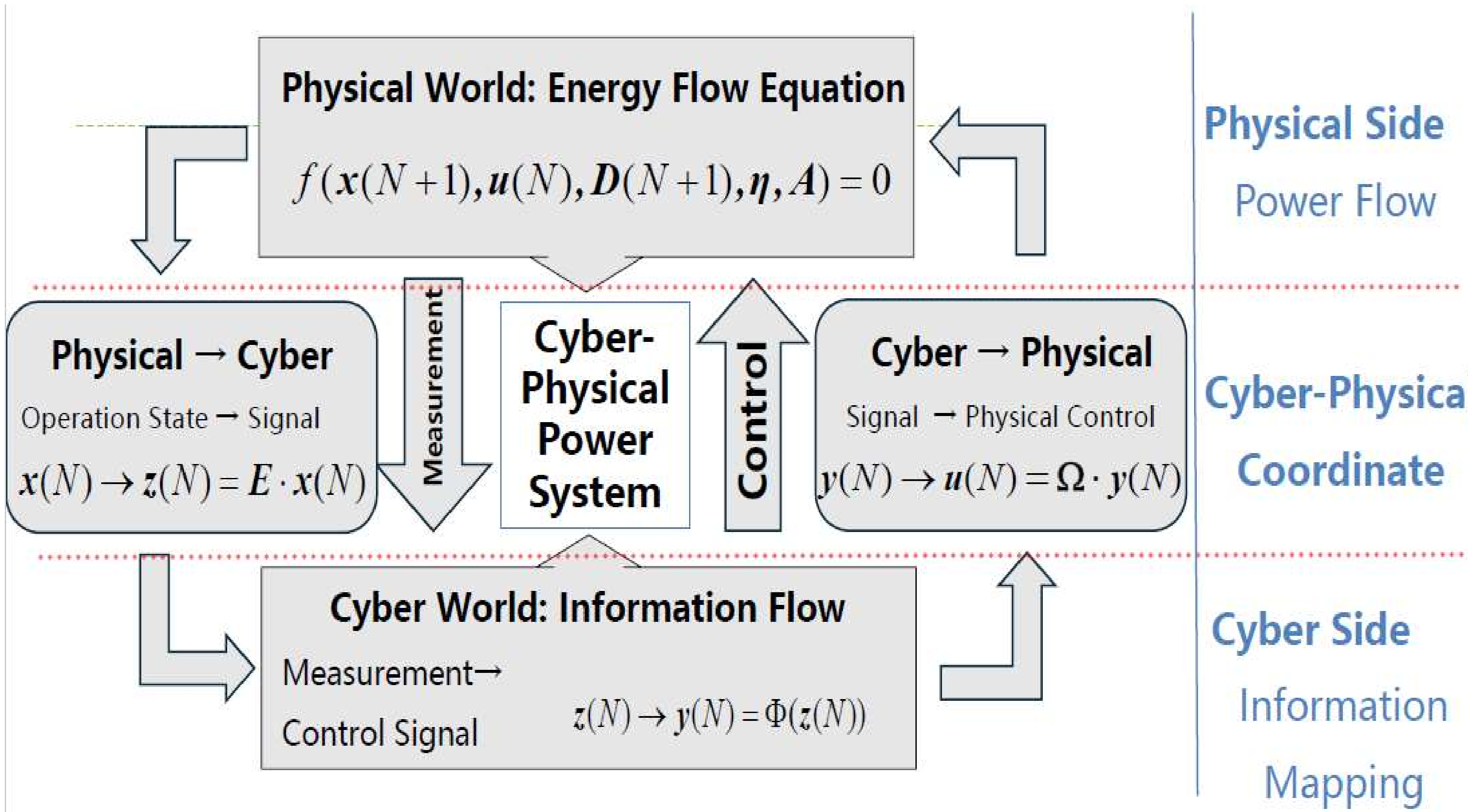
Cyber Contingencies and Power System



Challenge:

How to quantitatively evaluate the degree to which cyber-side contingencies influence the physical power system operation?

System State：Information Energy Flow Equation



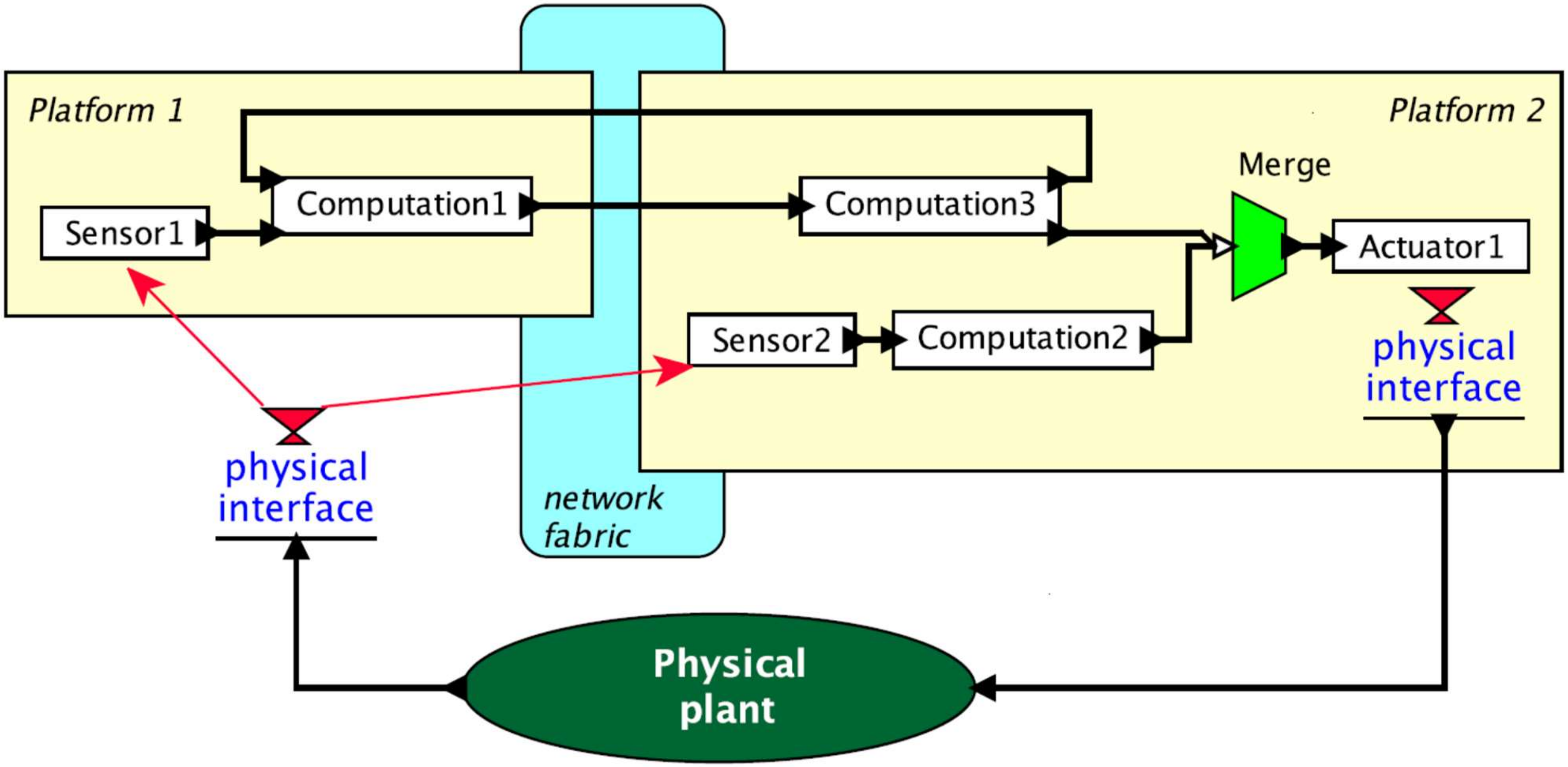


Figure 1.1: Example structure of a cyber-physical system.

Motivating Example

- The specific application is the Stanford testbed of autonomous rotorcraft for multi agent control **(STARMAC)**, developed by Claire Tomlin and colleagues as a cooperative effort at Stanford and Berkeley (Hoffmann et al., 2004).

- The STARMAC is a small quadrotor aircraft; it is shown in flight in Figure 1.2.

- Its primary purpose is to serve as a testbed for experimenting with multi-vehicle autonomous control techniques. The objective is to be able to have multiple vehicles cooperate on a common task.

Figure 1.2: The STARMAC quadrotor aircraft in flight (reproduced with permission).

How to go about designing and implementing cyber-physical systems?

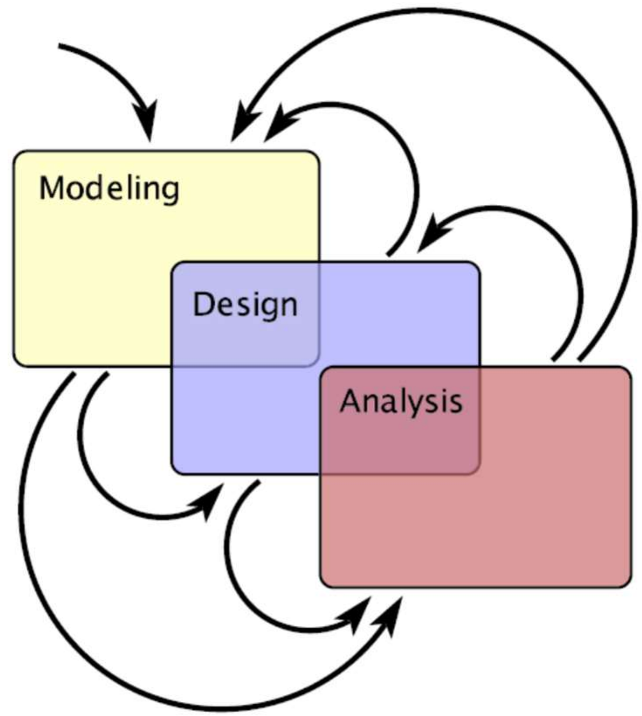


Figure 1.3: Creating embedded systems requires an iterative process of modeling, design, and analysis.

- Modeling is the process of gaining a deeper understanding of a system through imitation. Models imitate the system and reflect properties of the system. Models specify **what a system does**.

- Design is the structured creation of artifacts. It specifies **how a system does** what it does.

- Analysis is the process of gaining a deeper understanding of a system through dissection. It specifies **why a system does** what it does (or fails to do what a model says it should do).

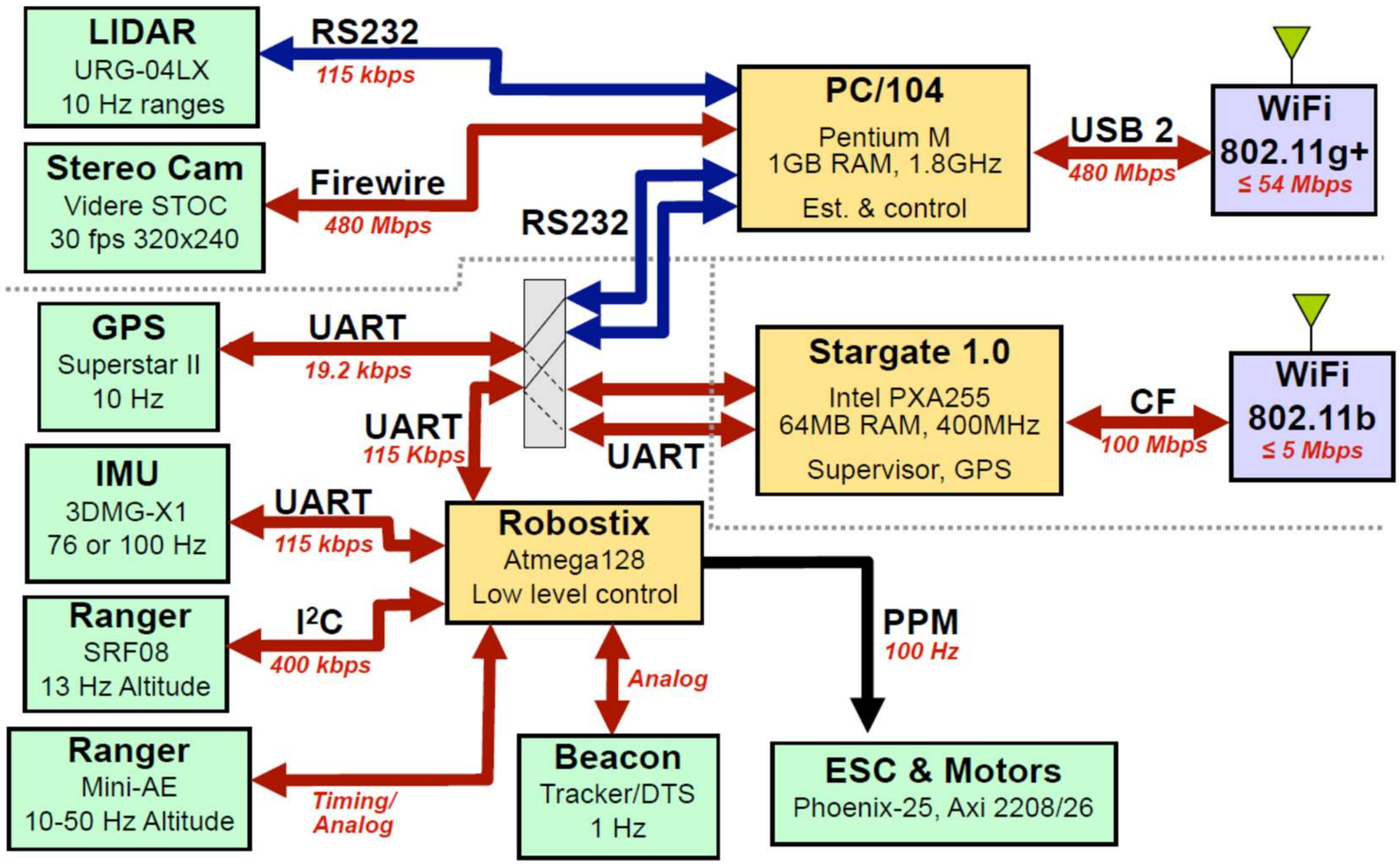


Figure 1.4: The STARMAC architecture (reproduced with permission).

Modeling

- A model of a physical system is a description of certain aspects of the system that is intended to yield insight into properties of the system.

- In this text, models have mathematical properties that enable systematic analysis.

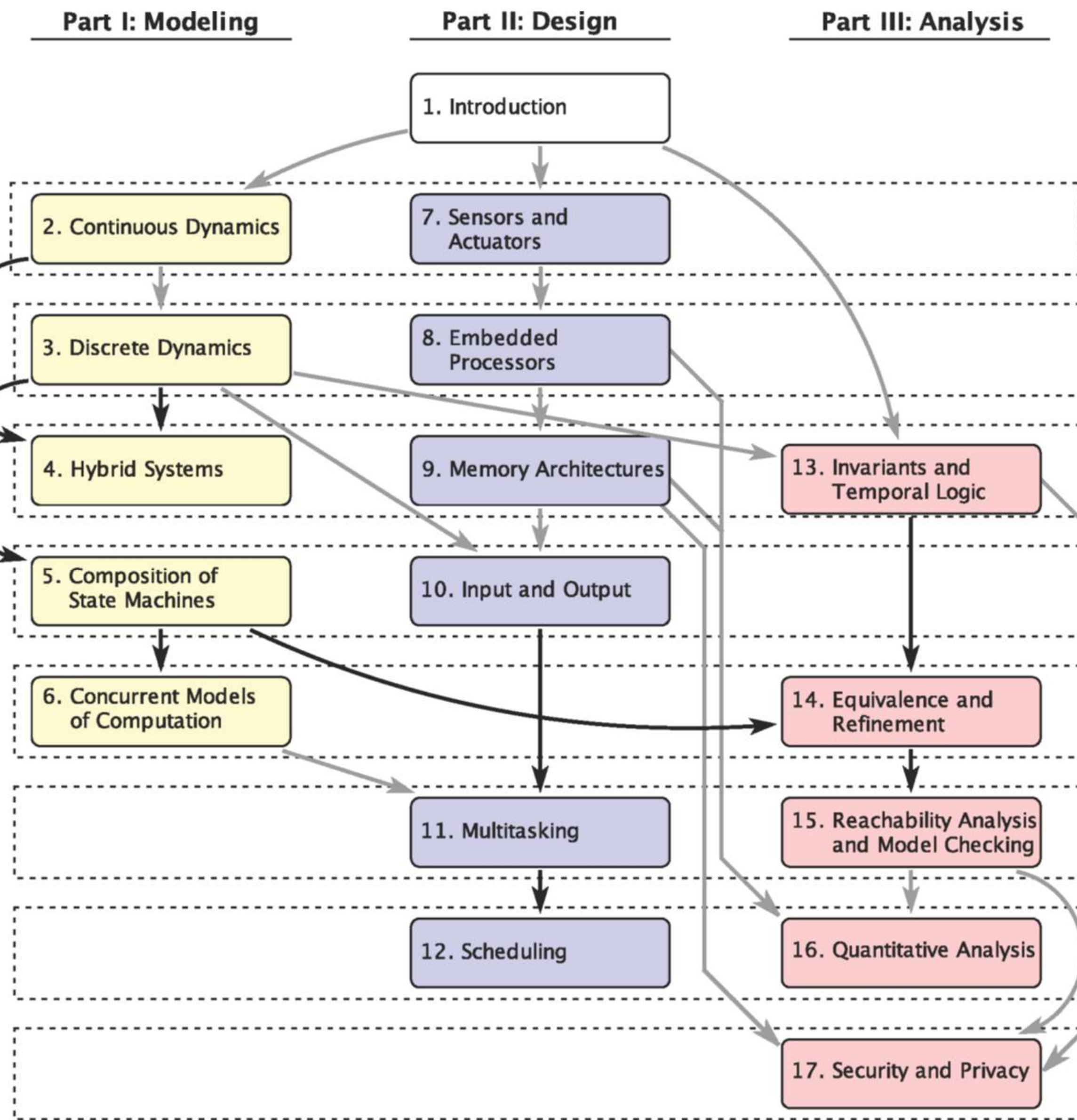
- The model imitates properties of the system, and hence yields insight into that system.

Design

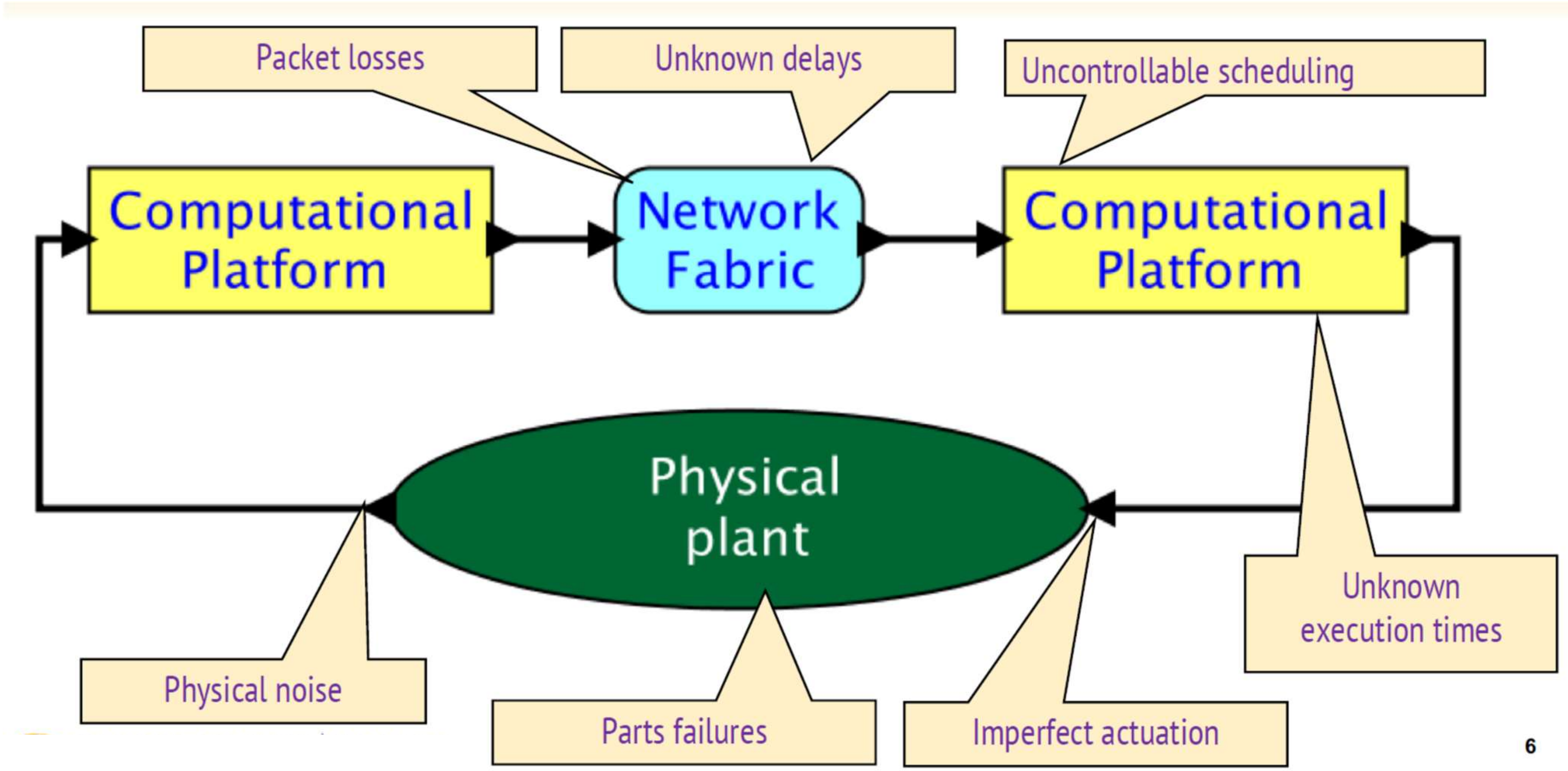
This part focuses on the design of embedded systems, with emphasis on the role they play within a CPS.

Analysis

The analysis part focuses on precise specifications of properties, on techniques for comparing specifications, and on techniques for analyzing specifications and the resulting designs.


Part I: Modeling
Part II: Design
Part III: Analysis
1. Introduction
2. Continuous Dynamics
3. Discrete Dynamics
4. Hybrid Systems
5. Composition of State Machines
6. Concurrent Models of Computation
7. Sensors and Actuators
8. Embedded Processors
9. Memory Architectures
10. Input and Output
11. Multitasking
12. Scheduling
13. Invariants and Temporal Logic
14. Equivalence and Refinement
15. Reachability Analysis and Model Checking
16. Quantitative Analysis
17. Security and Privacy

Schematic of a simple CPS - Uncertainties



Modeling Dynamic Behaviors

Continuous Dynamics

- Established techniques for modeling the dynamics of physical systems, with emphasis on their continuous behaviors

- Motion of mechanical parts can often be modeled using differential equations, or equivalently, integral equations. Such models really only work well for “smooth” motion (a concept that we can make more precise using notions of linearity, time invariance, and continuity).

- We begin with simple equations of motion, which provide a model of a system in the form of ordinary differential equations (ODEs).

- We then show how these ODEs can be represented in actor models, which include the class of models in popular modeling languages such as LabVIEW (from National Instruments) and Simulink (from The MathWorks, Inc.).

- We develop a simple example of a feedback control system that stabilizes an unstable system.

- Controllers for such systems are often realized using software, so such systems can serve as a canonical example of a cyber-physical system.

- The properties of the overall system emerge from properties of the cyber and physical parts.

Newtonian Mechanics

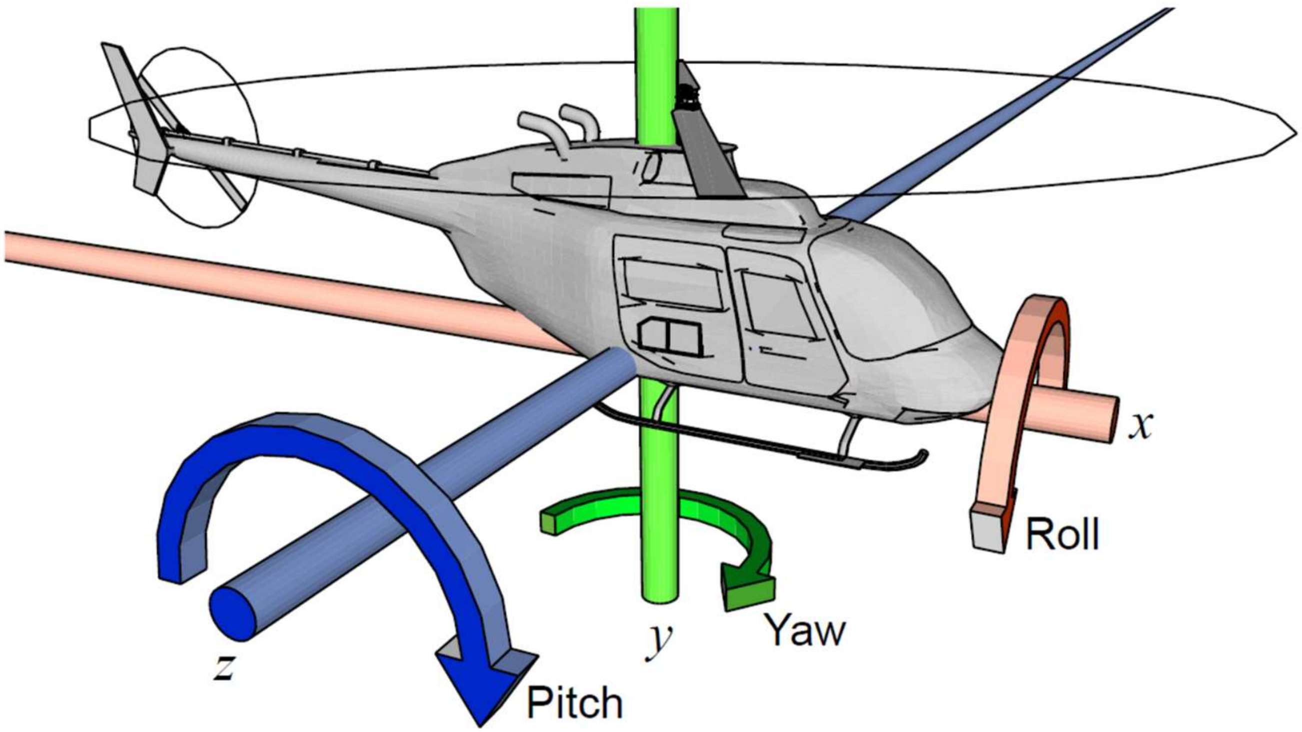


Figure 2.1: Modeling position with six degrees of freedom requires including pitch, roll, and yaw, in addition to position.

Motion in space of physical objects can be represented with **six degrees of freedom**, illustrated in Figure 2.1. Three of these represent position in three dimensional space, and three represent orientation in space. We assume three axes, x, y, and z, where by convention x is drawn increasing to the right, y is drawn increasing upwards, and z is drawn increasing out of the page. **Roll** θ_x is an angle of rotation around the x axis, where by convention an angle of 0 radians represents horizontally flat along the z axis (i.e., the angle is given relative to the z axis). **Yaw** θ_y is the rotation around the y axis, where by convention 0 radians represents pointing directly to the right (i.e., the angle is given relative to the x axis). **Pitch** θ_z is rotation around the z axis, where by convention 0 radians represents pointing horizontally (i.e., the angle is given relative to the x axis).

The position of an object in space, therefore, is represented by six functions of the form $f\colon \mathbb{R} \to \mathbb{R}$, where the domain represents time and the codomain represents either distance along an axis or angle relative to an axis.[1] Functions of this form are known as **continuous-time signals**.[2] These are often collected into vector-valued functions $\mathbf{x}\colon \mathbb{R} \to \mathbb{R}^3$ and $\theta\colon \mathbb{R} \to \mathbb{R}^3$, where $\mathbf{x}$ represents position, and θ represents orientation.

Changes in position or orientation are governed by **Newton's second law**, relating force with acceleration. Acceleration is the second derivative of position. Our first equation handles the position information,

$$\mathbf{F}(t) = M\ddot{\mathbf{x}}(t), \tag{2.1}$$

where $\mathbf{F}$ is the force vector in three directions, M is the mass of the object, and $\ddot{\mathbf{x}}$ is the second derivative of $\mathbf{x}$ with respect to time (i.e., the acceleration). Velocity is the integral of acceleration, given by

$$\forall\, t > 0, \quad \dot{\mathbf{x}}(t) \quad = \quad \dot{\mathbf{x}}(0) + \int_{0}^{t} \ddot{\mathbf{x}}(\tau) d\tau$$

where $\dot{\mathbf{x}}(0)$ is the initial velocity in three directions. Using (2.1), this becomes

$$\forall\, t > 0, \quad \dot{\mathbf{x}}(t) = \dot{\mathbf{x}}(0) + \frac{1}{M} \int_{0}^{t} \mathbf{F}(\tau) d\tau,$$

Position is the integral of velocity,

$$\begin{aligned}
\mathbf{x}(t) &= \mathbf{x}(0) + \int_0^t \dot{\mathbf{x}}(\tau)d\tau \\
&= \mathbf{x}(0) + t\dot{\mathbf{x}}(0) + \frac{1}{M}\int_0^t \int_0^\tau \mathbf{F}(\alpha)d\alpha d\tau,
\end{aligned}$$

where $\mathbf{x}(0)$ is the initial position. Using these equations, if you know the initial position and initial velocity of an object and the forces on the object in all three directions as a function of time, you can determine the acceleration, velocity, and position of the object at any time.

The versions of these equations of motion that affect orientation use **torque**, the rotational version of force. It is again a three-element vector as a function of time, representing the net rotational force on an object. It can be related to angular velocity in a manner similar to (2.1),

$$\mathbf{T}(t) = \frac{d}{dt}\left(\mathbf{I}(t)\dot{\theta}(t)\right), \tag{2.2}$$

where $\mathbf{T}$ is the torque vector in three axes and $\mathbf{I}(t)$ is the **moment of inertia tensor** of the object. The moment of inertia is a 3×3 matrix that depends on the geometry and orientation of the object. Intuitively, it represents the reluctance that an object has to spin around any axis as a function of its orientation along the three axes. If the object is spherical, for example, this reluctance is the same around all axes, so it reduces to a constant scalar I (or equivalently, to a diagonal matrix $\mathbf{I}$ with equal diagonal elements I). The equation then looks much more like (2.1),

$$\mathbf{T}(t) = I\ddot{\theta}(t). \tag{2.3}$$

To be explicit about the three dimensions, we might write (2.2) as

$$\begin{bmatrix} T_x(t) \\ T_y(t) \\ T_z(t) \end{bmatrix} = \frac{d}{dt}\left(\begin{bmatrix} I_{xx}(t) & I_{xy}(t) & I_{xz}(t) \\ I_{yx}(t) & I_{yy}(t) & I_{yz}(t) \\ I_{zx}(t) & I_{zy}(t) & I_{zz}(t) \end{bmatrix} \begin{bmatrix} \dot{\theta}_x(t) \\ \dot{\theta}_y(t) \\ \dot{\theta}_z(t) \end{bmatrix} \right) .$$

Here, for example, $T_y(t)$ is the net torque around the y axis (which would cause changes in yaw), $I_{yx}(t)$ is the inertia that determines how acceleration around the x axis is related to torque around the y axis.

Rotational velocity is the integral of acceleration,

$$\dot{\theta}(t) = \dot{\theta}(0) + \int_0^t \ddot{\theta}(\tau) d\tau,$$

where $\dot{\theta}(0)$ is the initial rotational velocity in three axes. For a spherical object, using (2.3), this becomes

$$\dot{\theta}(t) = \dot{\theta}(0) + \frac{1}{I}\int_{0}^{t} \mathbf{T}(\tau)d\tau.$$

Orientation is the integral of rotational velocity,

$$\begin{aligned}
\theta(t) &= \theta(0) + \int_0^t \dot{\theta}(\tau)d\tau \\
&= \theta(0) + t\dot{\theta}(0) + \frac{1}{I}\int_{0}^{t}\int_{0}^{\tau} \mathbf{T}(\alpha)d\alpha d\tau
\end{aligned}$$

where $\theta(0)$ is the initial orientation. Using these equations, if you know the initial orientation and initial rotational velocity of an object and the torques on the object in all three axes as a function of time, you can determine the rotational acceleration, velocity, and orientation of the object at any time.

Often, as we have done for a spherical object, we can simplify by reducing the number of dimensions that are considered. In general, such a simplification is called a **model-order reduction**. For example, if an object is a moving vehicle on a flat surface, there may be little reason to consider the y axis movement or the pitch or roll of the object.

Example 2.1: Consider a simple control problem that admits such reduction of dimensionality. A helicopter has two rotors, one above, which provides lift, and one on the tail. Without the rotor on the tail, the body of the helicopter would start to spin. The rotor on the tail counteracts that spin. Specifically, the force produced by the tail rotor must perfectly counter the torque produced by the main rotor, or the body will spin. Here we consider this role of the tail rotor independently from all other motion of the helicopter.

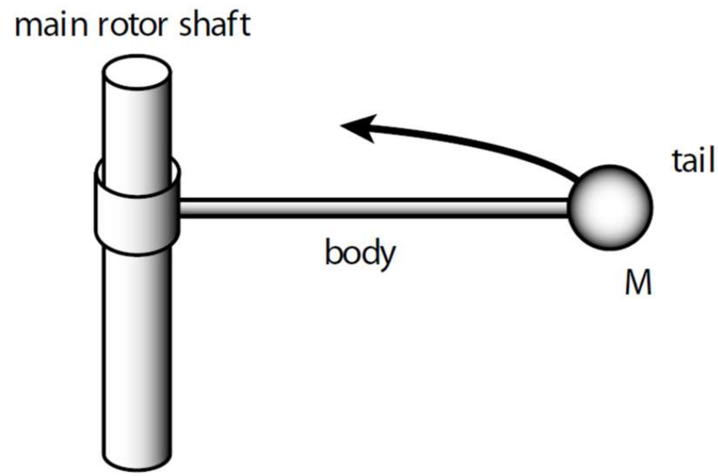


Figure 2.2: Simplified model of a helicopter.

We model the simplified helicopter by a system that takes as input a continuous-time signal T_y, the torque around the y axis (which causes changes in yaw). This torque is the net difference between the torque caused by the friction of the main rotor and that caused by the tail rotor. The output of our system will be the angular velocity $\dot{\theta}_y$ around the y axis. The dimensionally-reduced version of (2.2) can be written as

$$\ddot{\theta}_y(t) = T_y(t)/I_{yy}.$$

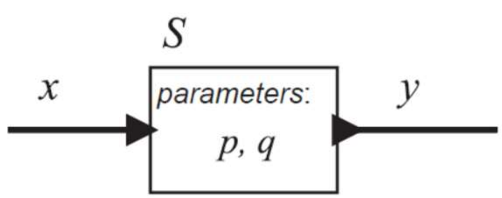
Integrating both sides, we get the output $\dot{\theta}$ as a function of the input T_y,

$$\dot{\theta}_y(t) = \dot{\theta}_y(0) + \frac{1}{I_{yy}} \int_0^t T_y(\tau)d\tau. \tag{2.4}$$

The critical observation about this example is that if we were to choose to model the helicopter by, say, letting $\mathbf{x}\colon \mathbb{R} \to \mathbb{R}^3$ represent the absolute position in space of the tail of the helicopter, we would end up with a far more complicated model. Designing the control system would also be much more difficult.

Actor Models

In the previous section, a model of a physical system is given by a differential or an integral equation that relates input signals (force or torque) to output signals (position, orientation, velocity, or rotational velocity). Such a physical system can be viewed as a component in a larger system. In particular, a **continuous-time system** (one that operates on continuous-time signals) may be modeled by a box with an input **port** and an output port as follows:

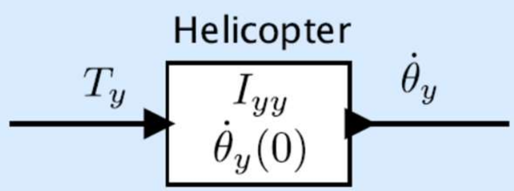


where the input signal x and the output signal y are functions of the form

$$x : \mathbb{R} \to \mathbb{R}, \quad y : \mathbb{R} \to \mathbb{R}.$$

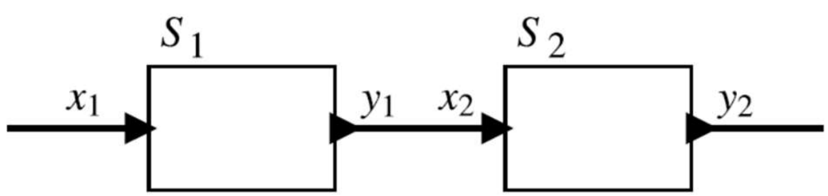
A box like that above, where the inputs are functions and the outputs are functions, is called an **actor**.

Example 2.2: The actor model for the helicopter of example 2.1 can be depicted as follows:



The input and output are both continuous-time functions. The parameters of the actor are the initial angular velocity $\dot{\theta}_y(0)$ and the moment of inertia I_{yy}. The function of the actor is defined by (2.4).

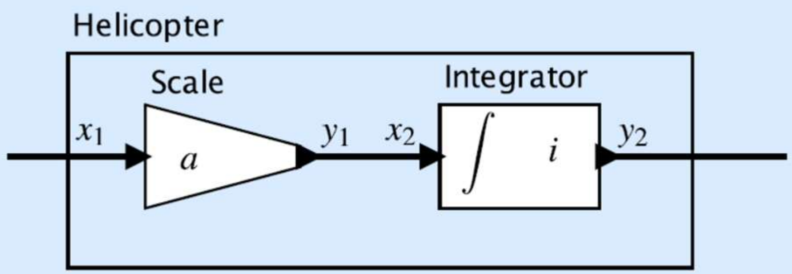
Actor models are composable. In particular, given two actors S_1 and S_2, we can form a **cascade composition** as follows:



In the diagram, the "wire" between the output of S_1 and the input of S_2 means precisely that $y_1 = x_2$, or more pedantically,

$$\forall\, t \in \mathbb{R}, \quad y_1(t) = x_2(t).$$

Example 2.3: The actor model for the helicopter can be represented as a cascade composition of two actors as follows:



The left actor represents a Scale actor parameterized by the constant a defined by

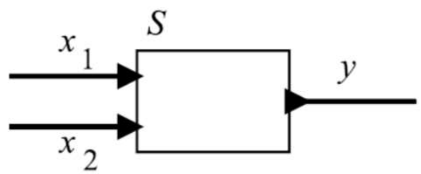
$$\forall\, t \in \mathbb{R}, \quad y_1(t) = a x_1(t). \tag{2.6}$$

More compactly, we can write $y_1 = a x_1$, where it is understood that the product of a scalar a and a function x_1 is interpreted as in (2.6). The right actor represents an integrator parameterized by the initial value i defined by

$$\forall\, t \in \mathbb{R}, \quad y_2(t) = i + \int_0^t x_2(\tau) d\tau.$$

If we give the parameter values $a = 1/I_{yy}$ and $i = \dot{\theta}_y(0)$, we see that this system represents (2.4) where the input $x_1 = T_y$ is torque and the output $y_2 = \dot{\theta}_y$ is angular velocity.

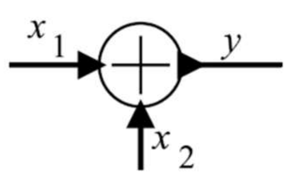
We can have actors that have multiple input signals and/or multiple output signals. These are represented similarly, as in the following example, which has two input signals and one output signal:



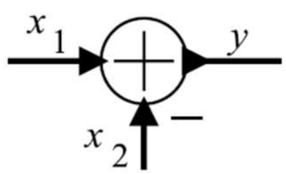
A particularly useful building block with this form is a signal **adder**, defined by

$$\forall\, t \in \mathbb{R}, \quad y(t) = x_1(t) + x_2(t).$$

This will often be represented by a custom icon as follows:



Sometimes, one of the inputs will be subtracted rather than added, in which case the icon is further customized with minus sign near that input, as below:



This actor represents a function $S\colon (\mathbb{R} \to \mathbb{R})^2 \to (\mathbb{R} \to \mathbb{R})$ given by

$$\forall\, t \in \mathbb{R},\ \forall\, x_1, x_2 \in (\mathbb{R} \to \mathbb{R}), \quad (S(x_1, x_2))(t) = y(t) = x_1(t) - x_2(t).$$

Notice the careful notation. $S(x_1, x_2)$ is a function in $\mathbb{R}^{\mathbb{R}}$. Hence, it can be evaluated at a $t \in \mathbb{R}$.

Properties of Systems

Causal Systems

Intuitively, a system is **causal** if its output depends only on current and past inputs. Making this notion precise is a bit tricky, however. We do this by first giving a notation for "current and past inputs." Consider a continuous-time signal $x\colon \mathbb{R} \to A$, for some set A. Let $x|_{t \le \tau}$ represent a function called the **restriction in time** that is only defined for times $t \le \tau$, and where it is defined, $x|_{t\le\tau}(t) = x(t)$. Hence if x is an input to a system, then $x|_{t\le\tau}$ is the "current and past inputs" at time τ.

Consider a continuous-time system $S\colon X \to Y$, where $X = A^{\mathbb{R}}$ and $Y = B^{\mathbb{R}}$ for some sets A and B. This system is causal if for all $x_1, x_2 \in X$ and $\tau \in \mathbb{R}$,

$$x_1|_{t\le\tau} = x_2|_{t\le\tau} \Rightarrow S(x_1)|_{t\le\tau} = S(x_2)|_{t\le\tau}$$

That is, the system is causal if for two possible inputs x_1 and x_2 that are identical up to (and including) time τ, the outputs are identical up to (and including) time τ. All systems we have considered so far are causal.

A system is **strictly causal** if for all $x_1, x_2 \in X$ and $\tau \in \mathbb{R}$,

$$x_1|_{t<\tau} = x_2|_{t<\tau} \Rightarrow S(x_1)|_{t\leq\tau} = S(x_2)|_{t\leq\tau}$$

That is, the system is strictly causal if for two possible inputs x_1 and x_2 that are identical up to (and *not* including) time τ, the outputs are identical up to (and including) time τ. The output at time t of a strictly causal system does not depend on its input at time t. It only depends on past inputs. A strictly causal system, of course, is also causal. The Integrator actor is strictly causal. The adder is not strictly causal, but it is causal. Strictly causal actors are useful for constructing feedback systems.

Memoryless Systems

Intuitively, a system has memory if the output depends not only on the current inputs, but also on past inputs (or future inputs, if the system is not causal). Consider a continuous-time system $S\colon X \to Y$, where $X = A^{\mathbb{R}}$ and $Y = B^{\mathbb{R}}$ for some sets A and B. Formally, this system is **memoryless** if there exists a function $f\colon A \to B$ such that for all $x \in X$,

$$(S(x))(t) = f(x(t))$$

for all $t \in \mathbb{R}$. That is, the output $(S(x))(t)$ at time t depends only on the input $x(t)$ at time t.

The Integrator considered above is not memoryless, but the adder is. Exercise 2 shows that if a system is strictly causal and memoryless then its output is constant for all inputs.

Linearity and Time Invariance

A system $S\colon X \to Y$, where X and Y are sets of signals, is linear if it satisfies the **superposition** property:

$$\forall\, x_1, x_2 \in X \text{ and } \forall\, a, b \in \mathbb{R}, \quad S(ax_1 + bx_2) = aS(x_1) + bS(x_2).$$

It is easy to see that the helicopter system defined in Example 2.1 is linear if and only if the initial angular velocity $\dot{\theta}_y(0) = 0$ (see Exercise 3).

More generally, it is easy to see that an integrator as defined in Example 2.3 is linear if and only if the initial value $i = 0$, that the Scale actor is always linear, and that the cascade of any two linear actors is linear. We can trivially extend the definition of linearity to actors with more than one input or output signal and then determine that the adder is also linear.

To define time invariance, we first define a specialized continuous-time actor called a **delay**. Let $D_\tau \colon X \to Y$, where X and Y are sets of continuous-time signals, be defined by

$$\forall\, x \in X \text{ and } \forall\, t \in \mathbb{R}, \quad (D_\tau(x))(t) = x(t-\tau). \tag{2.7}$$

Here, τ is a parameter of the delay actor. A system $S \colon X \to Y$ is time invariant if

$$\forall\, x \in X \text{ and } \forall\, \tau \in \mathbb{R}, \quad S(D_\tau(x)) = D_\tau(S(x)).$$

The helicopter system defined in Example 2.1 and (2.4) is not time invariant. A minor variant, however, is time invariant:

$$\dot{\theta}_y(t) = \frac{1}{I_{yy}} \int_{-\infty}^{t} T_y(\tau) d\tau.$$

This version does not allow for an initial angular rotation.

A **linear time-invariant system (LTI)** is a system that is both linear and time invariant. A major objective in modeling physical dynamics is to choose an LTI model whenever possible. If a reasonable approximation results in an LTI model, it is worth making this approximation. It is not always easy to determine whether the approximation is reasonable, or to find models for which the approximation is reasonable. It is often easy to construct models that are more complicated than they need to be (see Exercise 4).

Stability

A system is said to be **bounded-input bounded-output stable** (**BIBO stable** or just **stable**) if the output signal is bounded for all input signals that are bounded.

Consider a continuous-time system with input w and output v. The input is bounded if there is a real number $A < \infty$ such that $|w(t)| \leq A$ for all $t \in \mathbb{R}$. The output is bounded if there is a real number $B < \infty$ such that $|v(t)| \leq B$ for all $t \in \mathbb{R}$. The system is stable if for any input bounded by some A, there is some bound B on the output.

Example 2.4: It is now easy to see that the helicopter system developed in Example 2.1 is unstable. Let the input be $T_y = u$, where u is the **unit step**, given by

$$\forall\, t \in \mathbb{R}, \quad u(t) = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0 \end{cases} . \tag{2.8}$$

This means that prior to time zero, there is no torque applied to the system, and starting at time zero, we apply a torque of unit magnitude. This input is clearly bounded. It never exceeds one in magnitude. However, the output grows without bound.

In practice, a helicopter uses a feedback system to determine how much torque to apply at the tail rotor to keep the body of the helicopter straight. We study how to do that next.

Feedback Control

A system with **feedback** has directed cycles, where an output from an actor is fed back to affect an input of the same actor. An example of such a system is shown in Figure 2.3. Most control systems use feedback. They make measurements of an **error** (e in the figure), which is a discrepancy between desired behavior (ψ in the figure) and actual behavior ($\dot{\theta}_y$ in the figure), and use that measurement to correct the behavior. The error measurement is feedback, and the corresponding correction signal (T_y in the figure) should compensate to reduce future error. Note that the correction signal normally can only affect *future* errors, so a feedback system must normally include at least one strictly causal actor (the Helicopter in the figure) in every directed cycle.

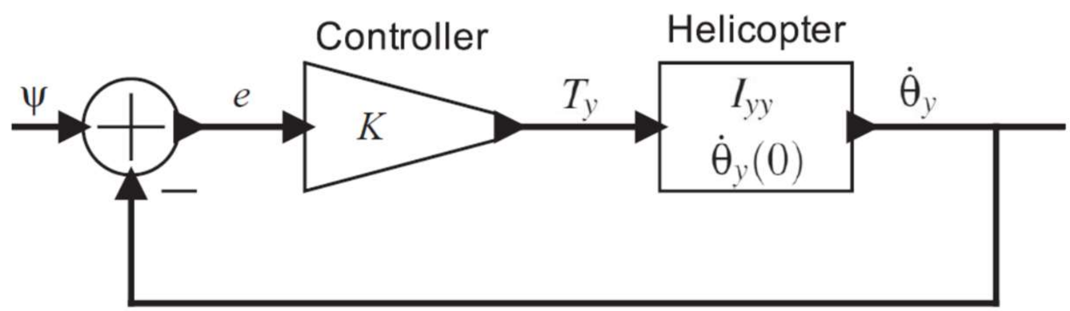


Figure 2.3: Proportional control system that stabilizes the helicopter.

Example 2.5: Recall that the helicopter model of Example 2.1 is not stable. We can stabilize it with a simple feedback control system, as shown in Figure 2.3. The input ψ to this system is a continuous-time system specifying the desired angular velocity. The **error signal** e represents the difference between the actual and the desired angular velocity. In the figure, the controller simply scales the error signal by a constant K, providing a control input to the helicopter. We use (2.4) to write

$$\dot{\theta}_y(t) = \dot{\theta}_y(0) + \frac{1}{I_{yy}} \int_0^t T_y(\tau) d\tau \tag{2.9}$$

$$= \dot{\theta}_y(0) + \frac{K}{I_{yy}} \int_0^t (\psi(\tau) - \dot{\theta}_y(\tau)) d\tau, \tag{2.10}$$

where we have used the facts (from the figure),

$$e(t) = \psi(t) - \dot{\theta}_y(t), \quad \text{and}$$

$$T_y(t) = Ke(t).$$

Equation (2.10) has $\dot{\theta}_y(t)$ on both sides, and therefore is not trivial to solve. The easiest solution technique uses Laplace transforms (see Lee and Varaiya (2011) Chapter 14). However, for our purposes here, we can use a more brute-force technique from calculus. To make this as simple as possible, we assume that $\psi(t) = 0$ for all t; i.e., we wish to control the helicopter simply to keep it from rotating at all. The desired angular velocity is zero. In this case, (2.10) simplifies to

$$\dot{\theta}_y(t) = \dot{\theta}_y(0) - \frac{K}{I_{yy}} \int_0^t \dot{\theta}_y(\tau) d\tau. \qquad (2.11)$$

Using the fact from calculus that, for $t \geq 0$,

$$\int_0^t a e^{a\tau} d\tau = e^{at} u(t) - 1,$$

where u is given by (2.8), we can infer that the solution to (2.11) is

$$\dot{\theta}_y(t) = \dot{\theta}_y(0) e^{-Kt/I_{yy}} u(t). \tag{2.12}$$

(Note that although it is easy to verify that this solution is correct, deriving the solution is not so easy. For this purpose, Laplace transforms provide a far better mechanism.)

We can see from (2.12) that the angular velocity approaches the desired angular velocity (zero) as t gets large as long as K is positive. For larger K, it will approach more quickly. For negative K, the system is unstable, and angular velocity will grow without bound.

The previous example illustrates a **proportional control** feedback loop. It is called this because the control signal is proportional to the error. We assumed a desired signal of zero. It is equally easy to assume that the helicopter is initially at rest (the angular velocity is zero) and then determine the behavior for a particular non-zero desired signal, as we do in the following example.

Example 2.6: Assume that helicopter is **initially at rest**, meaning that

$$\dot{\theta}(0) = 0,$$

and that the desired signal is

$$\psi(t) = au(t)$$

for some constant a. That is, we wish to control the helicopter to get it to rotate at a fixed rate.

We use (2.4) to write

$$
\begin{aligned}
\dot{\theta}_y(t) &= \frac{1}{I_{yy}} \int_0^t T_y(\tau) d\tau \\
&= \frac{K}{I_{yy}} \int_0^t (\psi(\tau) - \dot{\theta}_y(\tau)) d\tau \\
&= \frac{K}{I_{yy}} \int_0^t a d\tau - \frac{K}{I_{yy}} \int_0^t \dot{\theta}_y(\tau) d\tau \\
&= \frac{Kat}{I_{yy}} - \frac{K}{I_{yy}} \int_0^t \dot{\theta}_y(\tau) d\tau.
\end{aligned}
$$

Using the same (black magic) technique of inferring and then verifying the solution, we can see that the solution is

$$
\dot{\theta}_y(t) = au(t)(1 - e^{-Kt/I_{yy}}). \tag{2.13}
$$

Again, the angular velocity approaches the desired angular velocity as t gets large as long as K is positive. For larger K, it will approach more quickly. For negative K, the system is unstable, and angular velocity will grow without bound.

Note that the first term in the above solution is exactly the desired angular velocity. The second term is an error called the **tracking error**, that for this example asymptotically approaches zero.

The above example is somewhat unrealistic because we cannot independently control the *net* torque of the helicopter. In particular, the net torque T_y is the sum of the torque T_t due to the friction of the top rotor and the torque T_r due to the tail rotor,

$$\forall\, t \in \mathbb{R}, \quad T_y(t) = T_t(t) + T_r(t)\ .$$

T_t will be determined by the rotation required to maintain or achieve a desired altitude, quite independent of the rotation of the helicopter. Thus, we will actually need to design a control system that controls T_r and stabilizes the helicopter for any T_t (or, more precisely, any T_t within operating parameters). In the next example, we study how this changes the performance of the control system.

Example 2.7: In Figure 2.4(a), we have modified the helicopter model so that it has two inputs, T_t and T_r, the torque due to the top rotor and tail rotor respectively. The feedback control system is now controlling only T_r, and T_t is treated as an external (uncontrolled) input signal. How well will this control system behave?

Again, a full treatment of the subject is beyond the scope of this text, but we will study a specific example. Suppose that the torque due to the top rotor is given by

$$T_t = bu(t)$$

for some constant b. That is, at time zero, the top rotor starts spinning a constant velocity, and then holds that velocity. Suppose further that the helicopter is initially at rest. We can use the results of Example 2.6 to find the behavior of the system.

First, we transform the model into the equivalent model shown in Figure 2.4(b). This transformation simply relies on the algebraic fact that for any real numbers a_1, a_2, K,

$$Ka_1 + a_2 = K(a_1 + a_2/K).$$

We further transform the model to get the equivalent model shown in Figure 2.4(c), which has used the fact that addition is commutative. In Figure 2.4(c), we see that the portion of the model enclosed in the box is exactly the same as the control system analyzed in Example 2.6, shown in Figure 2.3. Thus, the same analysis as in Example 2.6 still applies. Suppose that desired angular rotation is

$$\psi(t) = 0.$$

Then the input to the original control system will be

$$x(t) = \psi(t) + T_t(t)/K = (b/K)u(t).$$

From (2.13), we see that the solution is

$$\dot{\theta}_y(t) = (b/K)u(t)(1 - e^{-Kt/I_{yy}}). \tag{2.14}$$

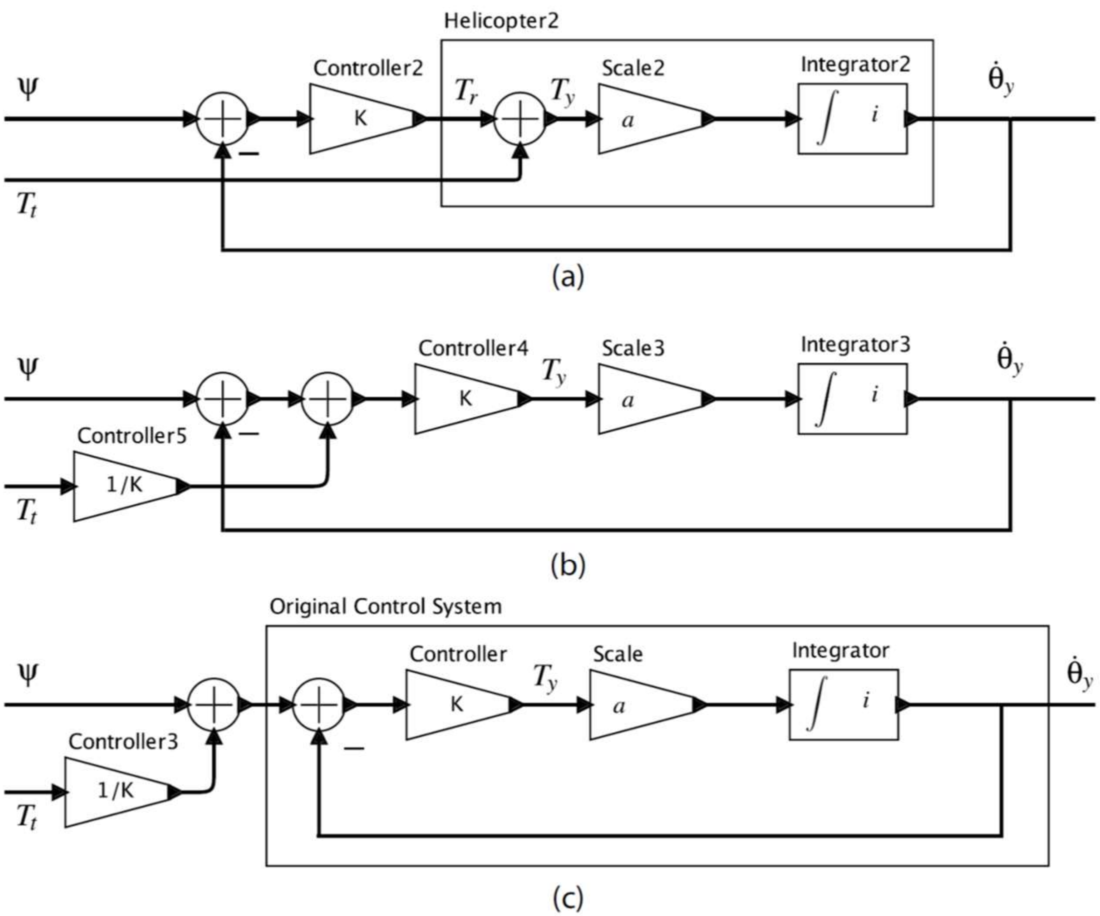


Figure 2.4: (a) Helicopter model with separately controlled torques for the top and tail rotors. (b) Transformation to an equivalent model (assuming $K > 0$). (c) Further transformation to an equivalent model that we can use to understand the behavior of the controller.

The desired angular rotation is zero, but the control system asymptotically approaches a non-zero angular rotation of b/K. This tracking error can be made arbitrarily small by increasing the control system feedback gain K, but with this controller design, it cannot be made to go to zero. An alternative controller design that yields an asymptotic tracking error of zero is studied in Exercise 6.

Summary

This chapter has described two distinct modeling techniques that describe physical dynamics. The first is ordinary differential equations, a venerable toolkit for engineers, and the second is actor models, a newer technique driven by software modeling and simulation tools. The two are closely related. This chapter has emphasized the relationship between these models, and the relationship of those models to the systems being modeled. These relationships, however, are quite a deep subject that we have barely touched upon. Our objective is to focus the attention of the reader on the fact that we may use multiple models for a system, and that models are distinct from the systems being modeled. The fidelity of a model (how well it approximates the system being modeled) is a strong factor in the success or failure of any engineering effort.